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Task-Aware Object Selection Using FPGA-Based Two-Stage Approximation

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Abstract

This work proposes a task-aware object detection system that selects the most suitable object in an image based on a given task. Unlike conventional methods that detect all objects, the system prioritises only task-relevant objects.

A task-aware scoring mechanism combines detection confidence with task relevance to filter irrelevant objects early. The selected candidates are then processed using an FPGA-based accelerator, which performs efficient similarity computation using a two-stage approach.

By combining intelligent filtering with hardware acceleration, the proposed system reduces computation, improves efficiency, and enables fast task-driven decision-making on edge platforms

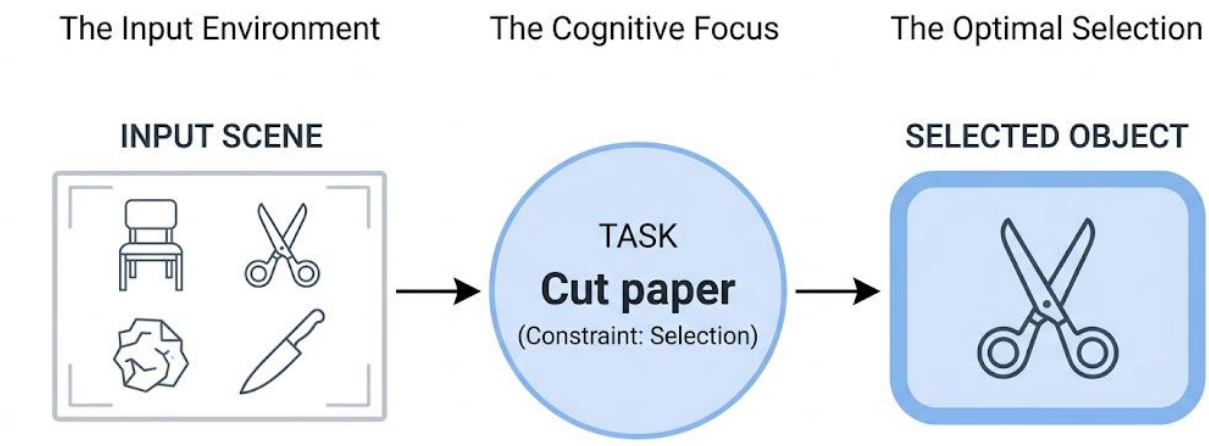


Fig 1: Basic input/output idea

I. Introduction

Recent advances in computer vision have enabled accurate detection and localization of multiple objects within images. However, conventional object detection systems are inherently task-agnostic, identifying all visible objects without considering their relevance to a specific task. This limits their effectiveness in real-world scenarios, where only contextually relevant objects matter.

To address this, the DVCon India 2026 Design Contest introduces task-aware object detection, where systems must select the most appropriate object based on a given task. For example, choosing between a cup, knife, or paper depends entirely on whether the task is “drink water” or “cut paper,” requiring contextual reasoning beyond traditional methods.

In this work, we propose a task-aware object selection system that combines visual detection with task-driven prioritization. The design integrates lightweight filtering on a VEGA processor with an FPGA-based accelerator to achieve efficient, low-latency decision-making.

This approach is optimized for edge deployment, enabling accurate and power-efficient task-aware perception for applications such as assistive robotics and autonomous systems.

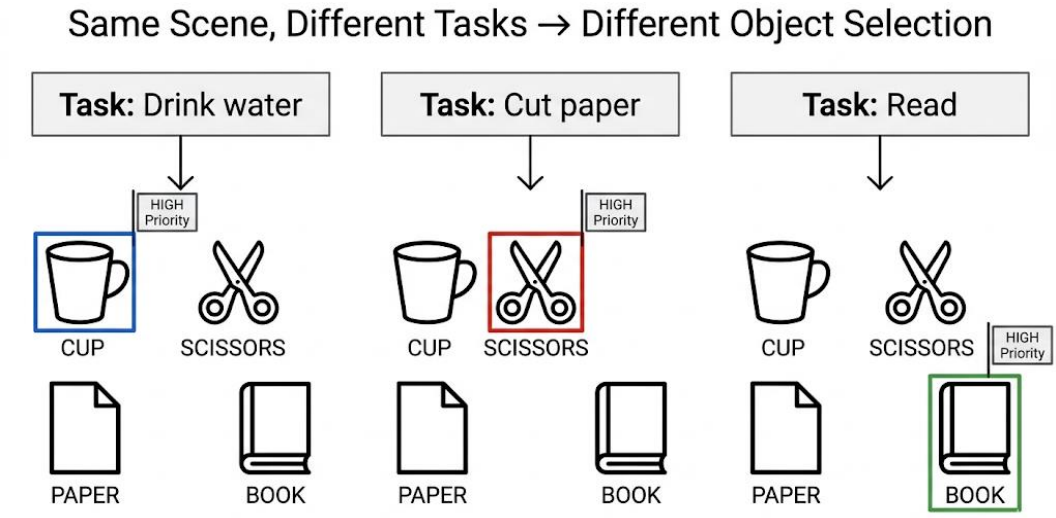


Fig 2: Tasks and image detection

II. Background Research

Traditional object detection systems such as YOLO and Faster R-CNN are widely used to detect and classify objects in images. These models assign confidence scores to detected objects based purely on visual features. However, they are inherently task-agnostic, meaning they do not consider the specific task while selecting objects.

Recent work, such as *"What Object Should I Use? – Task Driven Object Detection"*, introduces task-aware reasoning using Gated Graph Neural Networks (GGNN), where object relationships and task context are used to compute suitability scores.

However, existing approaches have the following limitations:

- High computational complexity due to graph-based reasoning
- Lack of optimisation for edge deployment

- No focus on hardware acceleration (FPGA-based systems)
- Increased latency and power consumption

The difference between conventional object detection and task-aware object selection is illustrated in Figure 1.

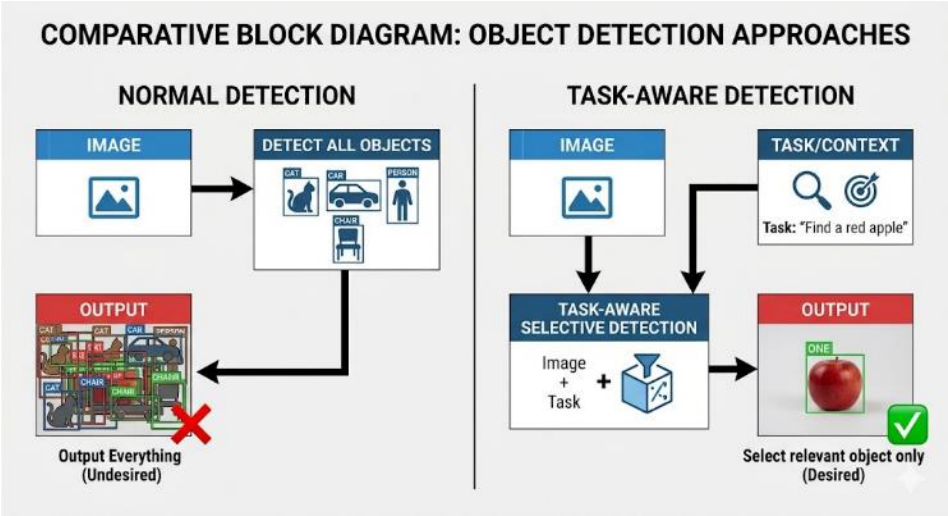


Fig 3: Comparative Block Diagram-Object Detection Approaches

The DVCon problem specifically requires a system that is:

- Task-aware
- Efficient for edge devices
- Optimised using the VEGA processor and the FPGA

Motivated by these gaps, our work focuses on:

- Lightweight task-aware prioritisation
- Early filtering of irrelevant objects
- Efficient hardware-accelerated similarity computation

III. Goal and Objectives

GOALS

- Enable task-aware object detection instead of conventional task-agnostic detection.
- Identify and select contextually relevant objects based on a given task.
- Improve decision-making accuracy in real-world scenarios.
- Support efficient deployment on edge systems with low latency and power usage.

OBJECTIVES

- Detect objects using a standard model (YOLO).
- Apply task-based filtering to reduce candidate objects.
- Rank objects using confidence and task relevance.
- Implement a fallback mechanism for robustness.
- Utilize VEGA + FPGA for efficient processing.
- Achieve low-latency, power-efficient performance.

IV. Design Process

i. Problem Statement

- Conventional object detection systems are inherently task-agnostic, identifying all objects within an image without considering the context of a specific task. This often leads to irrelevant outputs and limits their effectiveness in real-world, task-driven applications.
- Since object relevance depends on the intended task, there is a need for systems that incorporate task context to enable more accurate and meaningful object selection.

ii. Functional Specification

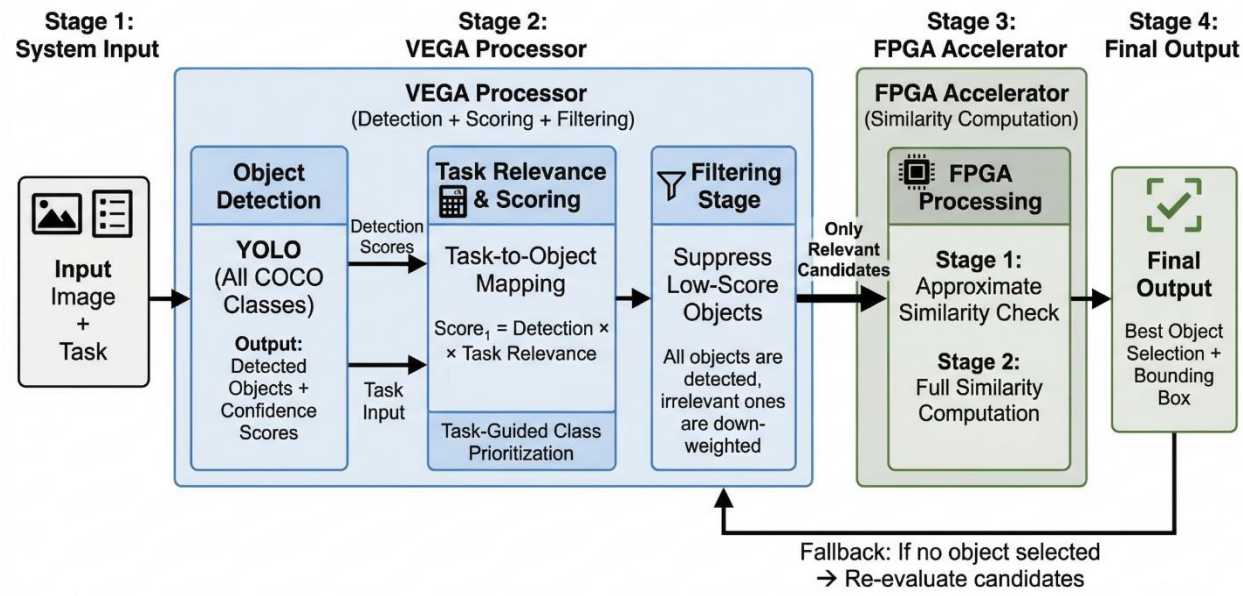


Fig 4: Overview of Task-Aware Object Selection System

The system performs task-aware object selection by combining object detection with task-based filtering and scoring to identify the most relevant object in a scene.

Inputs

- Input image containing multiple objects
- Task description in textual form

Outputs

- Selected object relevant to the given task
- Bounding box of the selected object

Task-Based Object Prioritization System (COCO Dataset)

Task	High Priority (0.9)	Medium Priority (0.7 / 0.5)	Default (0.2)
1. Step on something	Bench	Chair, Couch, Bed	All other COCO classes
2. Sit comfortably	Chair	Couch, Bed, Bench	All other COCO classes
3. Place flowers	Vase	Potted plant, Bowl	All other COCO classes
4. Get potatoes out of fire	Spoon	Fork, Knife, Bowl	All other COCO classes
5. Water plant	Potted plant	Bottle, Cup, Bowl	All other COCO classes
6. Get the lemon out of the tea	Spoon	Fork, Cup	All other COCO classes
7. Dig a hole	None (no suitable object available in COCO dataset)	Knife, Spoon, Bowl, Cup	All other COCO classes
8. Open a bottle of beer	Bottle	Knife, Spoon	All other COCO classes
9. Open parcel	Knife, Scissors	Fork, Spoon	All other COCO classes
10. Serve wine	Wine glass	Cup, Bottle	All other COCO classes
11. Pour sugar	Cup	Bowl, Spoon, Bottle	All other COCO classes
12. Smear butter	Knife	Spoon	All other COCO classes
13. Extinguish fire	None (no suitable object available in COCO dataset)	Bottle, Cup, Bowl	All other COCO classes
14. Pound carpet	Baseball bat	Tennis racket, Skateboard	All other COCO classes

Note: Priorities are assigned based on functional suitability and practical effectiveness of objects for each task. Default priority applies to all other classes not listed.

System Functions

- Detect objects in the image and obtain confidence scores using YOLO
- Map the input task to a predefined set of relevant object classes
- Filter detected objects based on task relevance
- Compute a task-aware score using detection confidence and task weight
- Select the object with the highest score
- Trigger a fallback mechanism when no suitable object is found

- Re-evaluate all detected objects and select the best available candidate

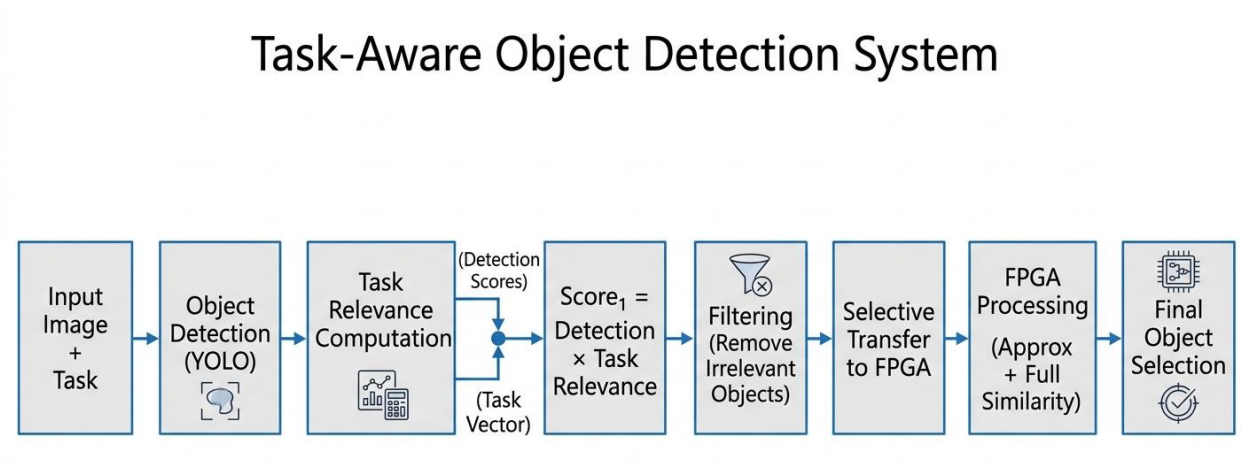


Fig 5: Flow Diagram of System

System Constraints

- Must operate efficiently on edge hardware (VEGA processor and FPGA)
- Should maintain low latency and power consumption
- Must minimize unnecessary computation through early filtering
- Should support real-time or near real-time performance

iii. Proposed Design

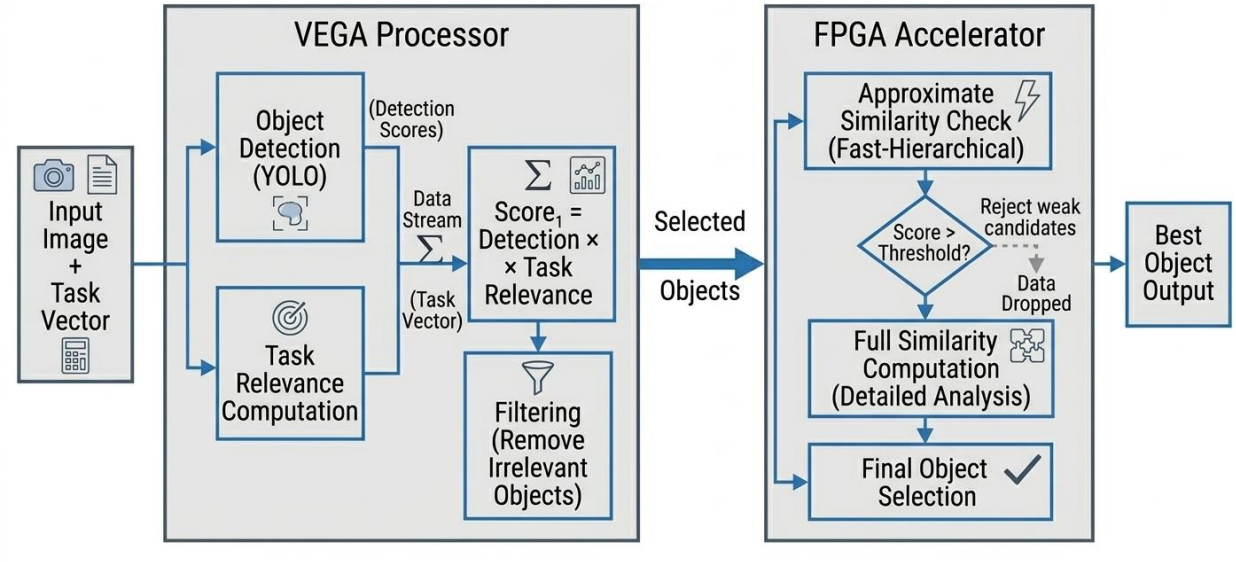


Fig 6: Task-Aware Object Detection System Architecture

The proposed system implements a task-aware object selection framework that combines object detection, task-based reasoning, and hardware acceleration for efficient edge deployment. The architecture follows a two-stage pipeline, with lightweight processing on the VEGA processor and compute-intensive operations on an FPGA.

The system takes an input image and a task description, and processes them sequentially to identify the most relevant object. The design focuses on early reduction of detected objects, followed by refined evaluation for accurate selection.

Stage 1: Task-Aware Scoring and Filtering (VEGA Processor)
Object detection is performed using YOLO to obtain detected objects along with their confidence scores and bounding boxes.

The input task is used to assign a task relevance score to each detected object. A task-aware score ($Score_1$) is computed as:

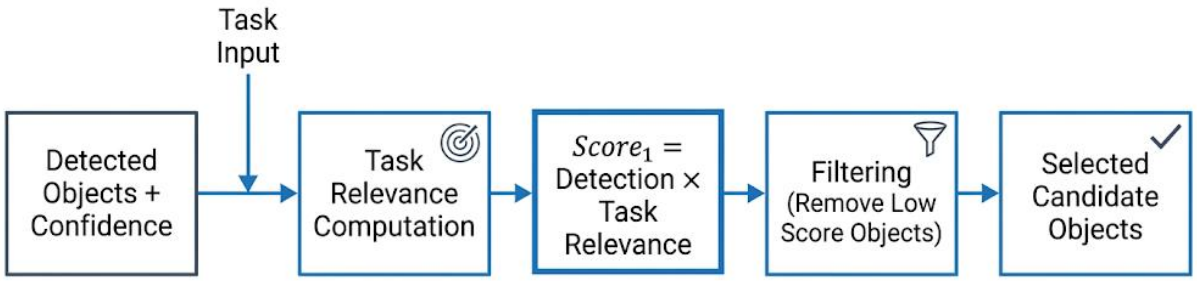


Fig 7: Stage1-Task Aware Scoring and Filtering

$$Score_1 = \text{Detection Confidence} \times \text{Task Relevance}$$

This score is used to prioritize objects and remove low-relevance detections. Only high-scoring objects are forwarded for further processing, reducing computational load.

Stage 2: FPGA-Based Similarity Evaluation

The selected objects are transferred to the FPGA for efficient evaluation using a hierarchical approach:

- **Stage 2.1: Fast Approximate Evaluation**
  Quickly eliminates weak objects using lightweight checks
- **Stage 2.2: Full Similarity Computation**
  Performs detailed evaluation on the remaining objects

The object with the highest similarity score ($Score_2$) is selected as the final output.

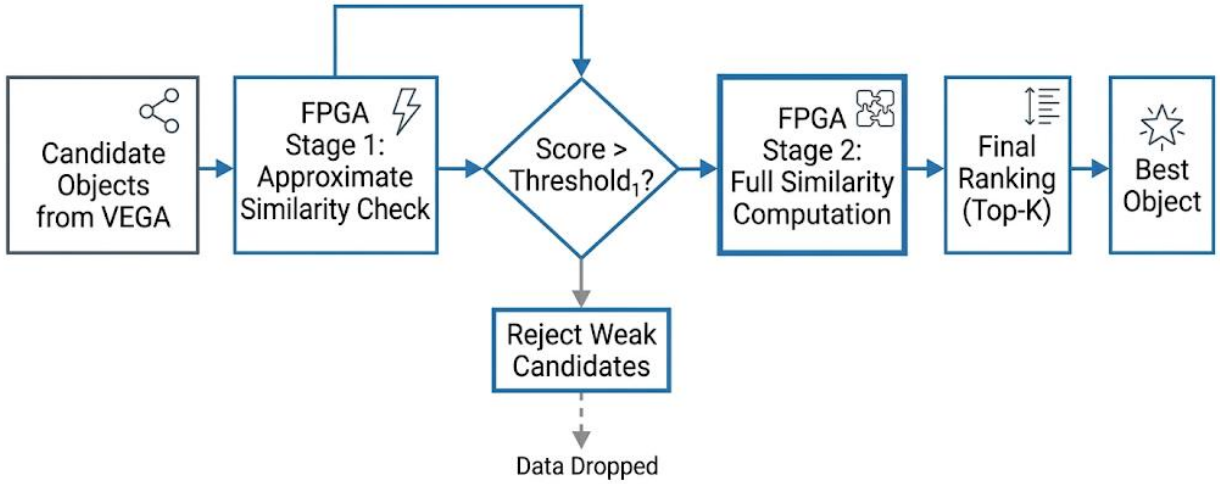


Fig 8: Stage2-FPGA Based evaluation

Fallback Mechanism

If no objects meet the Score$_1$ threshold or confidence is low, the system re-evaluates all detected objects. This ensures reliable output even in uncertain scenarios.

Data Flow and System Integration

- **VEGA processor** handles detection output processing, task mapping, and filtering
- **FPGA** performs similarity computation and final selection

Only selected objects are transferred, minimizing data movement and improving efficiency.

Design Highlights

- Task-aware prioritization using scoring
- Early filtering to reduce computation

- Selective VEGA–FPGA data transfer
- Hierarchical evaluation for efficiency
- Fallback mechanism for robustness
- Hardware–software co-design

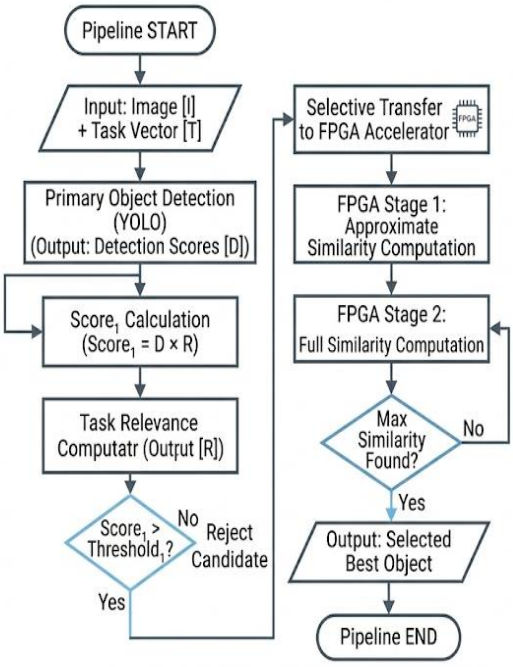


Fig 9: Pipelining Workflow

Design Advantages

The system reduces computational complexity by filtering early and offloading intensive tasks to the FPGA. This results in low latency, reduced power consumption, and efficient resource utilization, making it suitable for real-time edge applications.

Two-Stage Approximation

- The FPGA implements a two-stage approximation (coarse-to-fine) evaluation strategy to reduce computational overhead.

- **Stage 1: Approximate Evaluation**
  - Performs lightweight computation using reduced precision or partial features
  - Quickly eliminates low-relevance objects
  - Minimizes resource usage and latency
- **Stage 2: Refined Evaluation**
  - Applies full-precision computation on remaining objects
  - Ensures accurate comparison and final selection

  This approach reduces the number of high-cost operations, leading to improved latency, lower power consumption, and efficient resource utilization.

V. Results and Discussion

The proposed system demonstrates an efficient approach to task-aware object selection by integrating detection confidence with task relevance for early prioritization of meaningful objects. This enables reduction in unnecessary processing while improving the accuracy of object selection.

The implementation of selective filtering on the VEGA processor ensures that only relevant objects are forwarded for further evaluation, thereby minimizing data transfer and computational overhead. Additionally, the FPGA-based hierarchical accelerator enhances performance by performing early rejection of weak objects and restricting detailed computation to a reduced set of inputs.

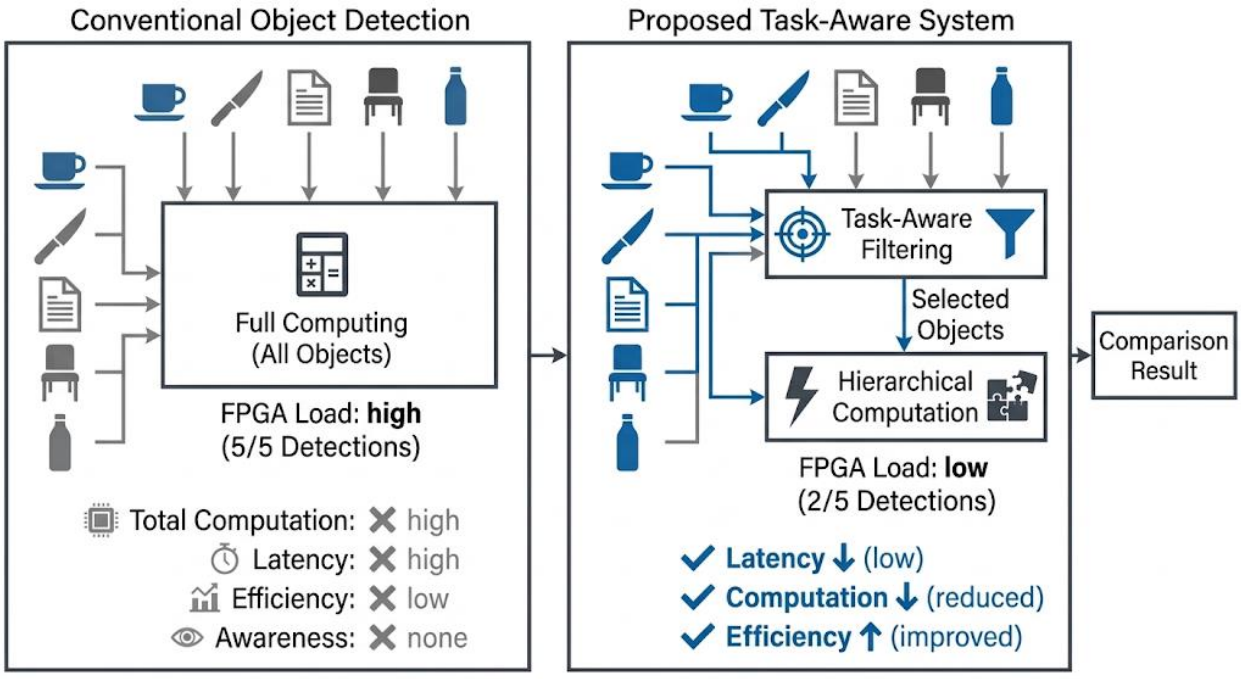


Fig 10: Comparison between conventional and proposed system

As a result, the system is expected to achieve lower latency, reduced power consumption, and improved decision quality compared to conventional task-agnostic object detection systems. Comprehensive implementation and quantitative evaluation will be carried out in subsequent stages to validate these performance improvements.

VI. Conclusion

This work presents a task-aware object selection system that combines task understanding with object detection to identify the most relevant object for a given task. By incorporating task-aware reweighting and early filtering, the system reduces unnecessary computation while improving

decision accuracy.

The use of a two-stage approximation-based FPGA accelerator,

along with a hardware–software co-design approach, enables efficient, low-latency operation suitable for edge deployment.

Overall, the system provides a practical and scalable solution for task-driven object selection. Future work will focus on refinement and full implementation on the VEGA and FPGA platform for detailed evaluation.

VII.References

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